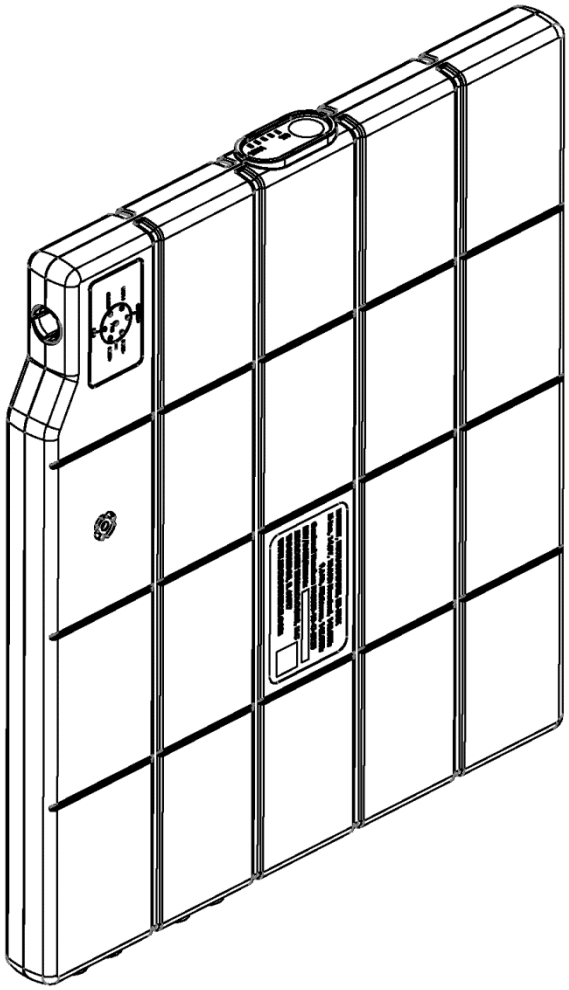
INCH-POUND
MIL-PRF-32383/4(CR)
6 March, 2013

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED, LITHIUM-ION, BB-25xx and BB-35xx

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32383



NOTES: See FIGURE 8 for full dimensions

FIGURE 1. M32383/4-3 (BB-25xx) Lithium Ion and M32383/4-4 (BB-35xx) Lithium Polymer Conformal Batteries

AMSC N/A FSC 6140
DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

Variances to MIL-PRF-32383 are shown as *italicized* text. Unless otherwise specified, paragraph numbers are in reference to MIL-PRF-32383. Note that new requirement and verification paragraphs are included in this document. Their numbering is based on MIL-PRF-32383.

APPLICABLE DOCUMENTS
The following additional documents are applicable to this battery

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-83133	*Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8), NATO F-35, and JP-8 + 100*

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	*Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment*

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

OTHER GOVERNMENT DOCUMENTS, DRAWINGS, AND PUBLICATIONS

ADS-37A-PRF	*Electromagnetic Environmental Effects (E^3) Performance and Verification Requirements*

(Copies of this document are available online at https://wwwproc.redstone.army.mil/acquisition/padds/web/W58RGZ05R0382/attach_exhib/ATTCH25ADS.pdf)

NON-GOVERNMENT PUBLICATIONS.

IEC 61000-4-5	*Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test*
IEC 61000-4-9	*Electromagnetic compatibility (EMC) – Part 4-9: Testing and measurement techniques Pulse magnetic field immunity test*

(Copies of this specification available online from International Electrotechnical Commission http://webstore.iec.ch)

ASTM G115-10	*Standard Guide for Measuring and Reporting Friction Coefficients*

(Copies of this specification are available online from ASTM International http://astm.org)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Type – III or Type IV (see 1.2 for classification)
PIN – M32383/4-3 or M32383/4-4 (see 1.2 for PIN)

Specification requirements and test methods: The following requirements and test methods of MIL-PRF-32383, identified therein as “when specified”, are applicable as indicated below:

Inspection	Requirement Paragraph	Test Method Paragraph	Applicability
Multilayer finish	3.4.4.1.3	Certification	Yes
Gold plating	*3.4.4.1.4*	*Certification*	*No*
Motor inrush current	3.5.5	4.7.2.12	Yes
Charge enable 1/	3.5.6	4.7.2.6	Yes
SMBus 1/	3.5.9	4.7.2.14; 4.7.2.15	Yes
Immersion, shallow 1/	3.6.6	4.7.3.8.1	Yes
Immersion, deep	*3.6.6*	*4.7.3.8.2*	*No*
Transit drop, normal	3.6.7	4.7.3.9.1	Group C only
Transit drop, severe	3.6.7	4.7.3.9.2	Yes
Drop while mated	*3.6.7*	*4.7.3.9.2*	*No*
Nail penetration, cell	3.7.1.2	4.7.4.4	Yes
Crush, cell	3.7.1.3	4.7.4.5	Yes
Projectile	3.7.2.6	4.7.4.11	Yes
Lithium Battery Safety Program (US Navy) Tests	3.7.2.7	4.7.5	Yes
Electromagnetic interference	3.8	4.7.7	Yes

1/ As modified herein

Details: The following are details specific to M32383/4-3 and M32383/4-4 including those referenced in MIL-PRF-32383 as “specified.”

1. *Certifications (3.2.1).*
   Add the following:

REQUIREMENT	*PARAGRAPH*
State-of-charge display characteristics	*3.5.8j*
Connector force disconnect	*3.4.4*
Smart Battery Identity	*3.5.6*
Overvoltage protection	*3.7.2.2.1*

2. Dimensions and weight (3.4.3)
   Width shall be no greater than 7.66 in
   Height shall be no greater than 8.70 in
   Weight shall be no greater than 2.6 lbs
   Depth shall be from 0.50 to 0.700 in

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

3. Battery connector and electrical interfaces (3.4.4)

Connector: GlenAir connector Part Number 807-348-01ZNU6 - 6SY or GlenAir 8070-1299-ZNU6-7DY or equivalent connector meeting the same interface and performance characteristics. The connector shall have a 15 +/- 5 pounds of force disconnect, certification is required. The primary (master) keyway shall be oriented downward, facing the charge contacts.

The connector shall provide a shell to shell shield impedance of not greater than 4 milliohms when mated with any of the mating connectors listed below.

Sockets 1, 2, and 3 shall be capable of carrying not less than 5 amps continuous each.

Socket 2 shall allow for mating with a longer pin connection that enables first mate – last break connections.

Mating Connector: GlenAir Part Number 807-309-06ZNU6-6PY – 6 pin straight connector, a 6 or 7 pin Right Angle connector (e.g. GlenAir 807-663-12 or 8071-1472-12 cable assembly), or equivalent connector.

Multilayer finish: Flat contacts shall be brass with multilayer finish

Gold Plating: N/A

Charge Contacts: The charge contacts shall be as shown in FIGURE 3 and FIGURE 8. Charge contacts shall be electrically protected against short circuit and shall not source battery current. All four contacts shall be recessed not greater than 0.03 inches so that if the battery was put on a flat metal surface, the contacts will not interact directly with the surface.

The battery shall be capable of being charged via both the main connector and the charge contacts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

Table 1: Battery Connector Electrical Interfaces

Connector Pinouts		
SOCKET	**DESCRIPTION**	**TERMINAL MARKINGS**
1	V+ Battery Voltage	Batt +
2	V- Battery Ground	Batt -
3	Charge +	Charge +
4	SMBus Data	Data
5	SMBus Clock	Clock
6	SMBus ID ('T-pin') -103AT Thermister to Ground	SB ID 1/
7 (If Present)	No Connection or 28V+ Battery Voltage	Blank / NC / Batt + 2/
Charge Contacts		
POSITION (Left to Right)	**DESCRIPTION**	**TITLE**
1	Charge Terminal, Positive	CHG +
2	Charger Terminal, Negative (Ground)	CHG -
3	SMBus Data	DATA
4	SMBus Clock	CLK

1/ Thermister pin IAW Smart Battery Data Specification Revision 1.1 paragraph 4.4.4

2/ if just 6 socket, blank, otherwise mark as configured.

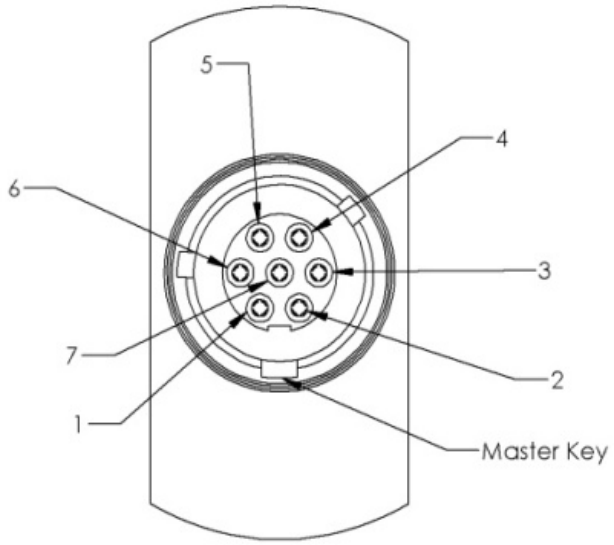


FIGURE 2. Connector

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

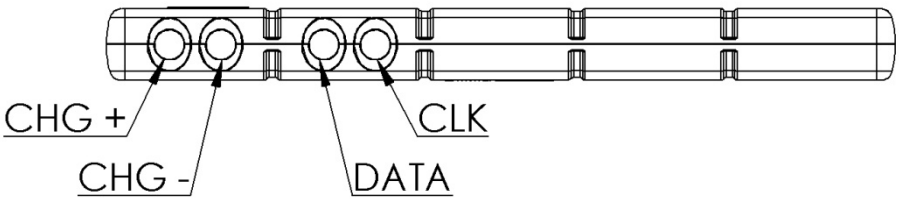


FIGURE 3. Charger Contact Assignments

4. Battery case (3.4.5)
Delete from MIL-PRF-32383 the following sentences:
"When plastics are used for the battery case, the material shall be classified in accordance with UL Standard 94, Test for Flammability of Plastic Materials for Parts in Devices and Appliances, except as otherwise noted herein. Acceptable ratings include: HB; V-0; V-1; V-2; 5VA; 5VB; VTM-0; VTM-1; and VTM-2."

And replace with the following sentences:
"The case material shall be classified in accordance with UL Standard 94, Test for Flammability of Plastic Materials for Parts in Devices and Appliances, except as otherwise noted herein. Acceptable ratings include: HB; V-0; V-1; 5VA; 5VB; VTM-0; and VTM-1."

5. Battery case vent (3.4.5.3)
Location is optional

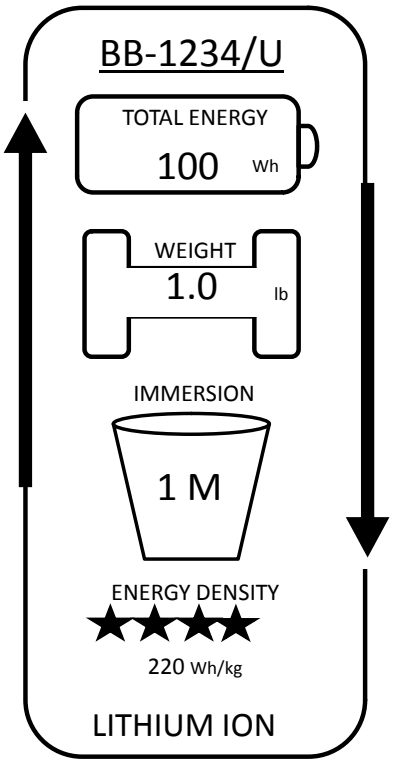
6. *Surface friction (3.4.5.6 – New paragraph)*
Batteries shall slide easily in and out of their pouch. When tested as specified in 4.7.1.8, batteries shall have a coefficient of static friction of not greater than 0.4 against a fabric containing meta-aramid, nylon and cotton in a blend ratio of 43/30/27 percent by weight.

7. Marking (3.4.6) and Identification Marking (3.4.6.1)
The location of the power interface marking shall be adjacent to the power connector and clearly identify socket assignments. Warning labels shall be not less than a 12 font, regular text shall not be less than a 10 point font and in accordance with MIL-STD-1472.

8. Instructions / notes (3.4.6.5)
The following charging instructions shall be marked on each battery:
Do not charge while below -4F (-20C) or above 122F (50C).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

9. Simplified battery label (3.4.6.6)
The simplified battery label location is optional. It shall be sized to remain legible. It may be split into multiple (2 maximum) sections.
The Simplified battery label shall be modified as shown below and display a bucket indicating the 2 hour immersion rating in meters.



For more information see
http://battery.army.mil

Note: Battery nomenclature and values are fictitious

FIGURE 4. Simplified battery label

10. Battery condition for shipping (3.4.8)
The instruction card shall at a minimum contain operation, storage, handling, maintenance instructions (including long term shelf maintenance requirements), MSDS#, Class of shipping and associated shipping markings IAW UN transportation regulations, and warranty information. *Batteries shall be between 40% and 60%State of Charge for shipping.*

11. Battery voltage (3.5.2)
Battery open circuit voltage: Min 10.0 volts, Max 17 volts

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

12. Capacity (3.5.3)

Minimum final voltage: 10 volts
Each battery, discharged at the specified rate to minimal final voltage, shall have the following minimum capacity:

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate	Minimum Capacity
Capacity Discharge	3.5.3	4.7.2.3	See detail 30	142 WH
Cycle life (224th cycle)	3.5.4	4.7.2.4	See detail 32	120 WH
High rate discharge	3.5.3	4.7.2.5	5 A	9.6 AH
Low temperature discharge	3.5.3	4.7.2.7	18 W	75 WH
High temperature discharge	3.5.3	4.7.2.8	18 W	140 WH
Motor inrush current	3.5.5	4.7.2.12	See detail 13	142 WH
Charge acceptance	3.5.10.1	4.7.2.9	18 W	142 WH
Retention of charge	3.5.6	4.7.2.10	18 W	135 WH
Pulse discharge	3.5.3	4.7.2.11	See detail 22	7.0 AH
Extreme low temperature discharge	3.6.1	4.7.3.2	18 W	75 WH
Extreme high temperature discharge	3.6.1	4.7.3.3	18 W	130 WH
Thermal shock (post-test)	3.6.3	4.7.3.5	18 W	130 WH
Mechanical shock (post-test)	3.6.3	4.7.3.6	18 W	135 WH
Vibration (during test)	3.6.5	4.7.3.7	1 A	142 WH
Vibration - loose cargo	*3.6.5*	*4.7.3.7*	*18W*	*142WH*
Battery storage life	3.6.8	4.7.3.10	See detail 30	100 WH
Impact	*3.6.9*	*4.7.3.11*	*18W*	*142 WH*
Conformability	*3.6.10*	*4.7.3.12*	*18W*	*142 WH*
Explosive Decompression	*3.6.11*	*4.7.3.13*	*18W*	*142 WH*
Electrostatic discharge	*3.6.13*	*4.7.3.17*		
Pulse magnetic field immunity	*3.6.14*	*4.7.3.18*	*18W*	*142 WH*
Salt Fog	*3.6.15*	*4.7.3.16*	*18W*	*142 WH*
Overcharge/electric leakage (post-test)	3.7.2.2	4.7.4.7	See detail 47	140 WH
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	18 W	135 WH
High temperature temporary cut off	3.7.2.4	4.7.4.9	18 W	105 WH
Electromagnetic interference	*3.8*	*4.7.7*	*18W*	*142 WH*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

13. Motor inrush (3.5.5).

The discharge profile shall be no less than as follows. (Note: this pulse is modeled after the end item and not a motor)

Step	Start time (second:millisecond:microsecond)	End time (second:millisecond:microsecond)	Current (A)
1	00:000:000	00:000:**915**	33
2	00:000:**915**	00:**025:000**	25
3	00:**025:000**	Until Cutoff Voltage	2

14. Charge enable (3.5.6)

Delete 3.5.6 from MIL-PRF-32383 in its entirety and replace with the following:

3.5.6 Charge enable. The charge enable socket, or Smart Battery Identity (SBID) socket (see figure 2), shall comply with the following. Certification is required:

a. Smart Battery Data (SBData) Specification v1.1 – accumulating battery information to support active battery management.

b. SMBus v1.1 - for the communication of SBData to a System Management Bus (SMBus) compliant host device to support active battery management.

Notes:

i. SMBus to enable and disable sleep mode

ii. PEC – Packet Error Checking Supported to improve reliability and communication robustness

iii. Data and clock series resistance output of < 300 ohms

iv. VCELL1-4, cell or cell string voltage readings implemented

v. Shall accommodate both Level 2 and Level 3 chargers

15. State-of-charge indicator (SOCI) (3.5.8).

Delete 3.5.8 from MIL-PRF-32383 in its entirety and replace with the following:

3.5.8 State-of-charge indicator (SOCI) display characteristics. The display shall have the following characteristics:

a. The SOCI shall be located on the top central edge of the battery pack. No part or component of the device shall protrude beyond the surface of the battery and not be protected from inadvertent operation. It shall be operable via depression with winter gloves or a mitten on.

b. The load applied by the State of Charge circuitry on the battery shall be not greater than 50 micro-amperes in the non-operating mode.

c. The device shall be designed to compensate for usage temperature and discharge rate and shall monitor and adjust for changes in both during battery use.

d. In the event that an open-circuit condition should occur in a battery (within the pack prior to the fuel gage) – ground or V+, the SOCI shall indicate no remaining capacity.

e. The SOCI shall display five (5) distinct ranges of remaining capacity in the battery, one range for each of five LEDs. The five ranges are listed in para 3.5.8i below.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

f. *The SOCI shall consist of a single row of green Light Emitting Diodes (LEDs) of equal size and an actuator button. The LEDs and the actuator shall be located in accordance with the applicable specification sheet. The output of each Light Emitting Diode (LED) at full intensity, measured in a direction normal to the State of Charge Indicator display, shall be 1.0 to 10.0 millicandelas. The LEDs shall be capable of providing the correct state of charge indication in a temperature range of −30 to 55ºC and be capable of surviving temperature extremes of −40ºC to 93ºC. Certification to the full intensity output and storage temperature range requirements is required.*

g. *The SOCI display shall normally be in the 'off' mode and shall require a push-button actuator to activate the LED display. The push-button shall be capable of actuation by finger. It shall be protected from inadvertent actuation. The words "PRESS FOR REMAINING CAPACITY" shall appear in close proximity to the actuator or the actuator shall be clearly visible to ensure its function is known. The push-button actuator shall comply with the following displacement and resistance requirements:*

	DISPLACEMENT	*RESISTANCE*
Not less than	*0.008 in.*	*0.25 lbs*
Not greater than	*0.050 in.*	*5.50 lbs*

h. *The LEDs shall display and be marked as to where the 0% and the 100% capacity levels are. At full intensity, the LEDs shall only illuminate the state of charge level appropriate to its usage level when viewed from a direction normal to the indicator (i.e., no light bleed-through to levels/spaces meant for higher states of charge). The LEDs shall remain at full intensity for not less than 3 seconds and shall self-extinguish in not greater than 7 seconds of actuation when the actuator is held down. The LEDs shall extinguish any time the actuator is released in not greater than 7 seconds.*

i. *During testing in accordance with 4.7.2.14, the SOCI shall indicate the correct state of charge at each stage of discharge. The actuator shall comply with displacement and resistance values stated above in 3.5.8g. Time to full light intensity shall be not greater than two seconds. Time for the LEDs to remain at full intensity shall be not less than 3 seconds; Time until the lights extinguish while the actuator is depressed shall be not greater than 7 seconds. The indicator shall not illuminate when the battery face including the actuator is pressed against a transparent flat surface. SOC indications obtained during testing shall comply with the table below. After discharge to final voltage, the indicator shall indicate no remaining capacity. Refer to Table 6.*

SOC Thresholds	*Test Points*	*Required SOC Readings*
100%	*90%*	*5 LEDs*
80%	*70%*	*4 LEDs*
60%	*50%*	*3 LEDs*
40%	*30%*	*2 LEDs*
20%	*10%*	*1 LEDs*
10%	*5%*	*1 LED Flash*
	0%	*0 LEDs*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

j. In accordance with the state of charge requirements herein, the following require certification (see 3.2.1).

Load limit (not greater than 150 Mili-amperes when LEDs are on)

LED full intensity of 1.0-10.0 milli-candelas each in a direction normal to display

Storage temperature range capability

16. *SMBus (3.5.9)*
SMBus shall use 5V logic circuitry. SMBus circuitry shall wake up in not greater than 210 seconds from a suspend or hibernate state, when not greater than 5.5V limited to not greater than 100 mA is applied to the charge contacts. Voltages greater than 17.2 V shall not wake up the battery or charge the battery. Certification of wake up and charge voltages is required.

17. *Impact (3.6.9 – New paragraph)*
Batteries shall be capable of being impacted without sustaining physical or electrical damage. When tested as specified in 4.7.3.11, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6 (with the exception of defects in 106 of TABLE VIII).

18. *Conformability (3.6.10 – New paragraph)*
Batteries shall be capable of flexing without sustaining physical or electrical damage. When tested as specified in 4.7.3.12 batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

19. *Explosive Decompression (3.6.11 – New paragraph)*
Batteries shall be capable of rapid changes in atmospheric pressure without sustaining physical or electrical damage. When tested as specified in 4.7.3.14, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

20. *Solar Radiation (3.6.12 – New paragraph)*
Batteries shall be capable of withstanding prolonged exposure to sunlight without sustaining physical or electrical damage. When tested as specified in 4.7.3.15 batteries shall meet the requirements of 3.6.

21. *Electrostatic discharge (ESD) (3.6.13 – New paragraph).*
Batteries shall withstand ESD to ±8kV, contact discharge and ±15kV air discharge on any surface, panel component (including vent) or connector, including exposed pins, in both operating and non-operating mode. When tested as specified in 4.7.3.17 batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

22. *Pulse magnetic field immunity (3.6.14 – New paragraph)*
Batteries shall withstand pulse magnetic fields. When tested as specified in 4.7.3.14 batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

23. *Salt fog (3.6.15 – New paragraph)*
Batteries shall withstand a salt fog atmosphere. When tested as specified in 4.7.3.16 batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

24. *Overvoltage protection (3.7.2.2.1 – New Paragraph)*
Batteries shall have three levels of overvoltage protection between each set of charge contacts (both primary connector and charge contacts) and the battery pack that shall withstand not less than 50V continuous. There shall be not less than two different circuit types and a single mode of failure shall not allow the cells to see an overvoltage condition. Certification is required.

25. *Bullet penetration (3.7.2.8 – New paragraph)*
Batteries shall not injure the user when struck by bullets or shrapnel or present unsafe conditions. When tested as specified in 4.7.4.12 the battery shall not cause an immediate flammable reaction and it shall allow not less than a two minute delay for safe removal from the Warfighter's vest post impact.

26. Electromagnetic compatibility (3.8):
Delete 3.8 from MIL-PRF-32383 in its entirety and replace with the following:

The battery shall be appropriately filtered and shielded to comply with the applicable emissions and susceptibility requirements of MIL-STD-461 Table V, Aircraft, Army, Including Flight Line and as tailored herein. When tested as specified in 4.7.7 the batteries and system shall function properly.
The battery shall exceed MIL-STD-461 EMI parameters by 20 dB for Communications frequencies (30 to 520 MHz; 762 to 870 MHz); SATCOM (1-2.5 GHz), and GPS Navigation L1 (1575.42 MHz) and L2 (1227.60 MHz) frequency bands.

27. *First article inspection (4.4)*
Delete from MIL-PRF-32383 the following sentences:
"a. Group 1 (GR1) – Functional tests, shall contain 12 samples
b. Group 2 (GR2) – Electromagnetic interference tests, when specified (see 3.1) shall contain five samples
c. Group 3 (GR3) – Interchangeability tests, shall contain eight samples
d. Group 4 (GR4) – Lithium safety tests, when specified (see 3.1) shall contain 15 samples"

And replace with the following sentence:
"a. Group 1 (GR1) – Functional tests, shall contain 16 samples
b. Group 2 (GR2) – Electromagnetic interference tests, when specified (see 3.1) shall contain two samples
c. Group 3 (GR3) – Interchangeability tests, shall contain eight samples
d. Group 4 (GR4) – Lithium safety tests, when specified (see 3.1) shall contain 15 samples"

28. Constant potential, current limited (4.6.1).
A constant potential of 16.8 volts shall be applied with current limited to 5.0 amperes, to a charge cut off of 750 milliamperes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

29. Alternate charging (4.6.3).
The following charger may be used whenever the general specification allows use of alternate charging:

	PART NUMBER	NSN	CAGEC
Charger (UltraLife)	CH0008	6130-01-545-1983	0UU59
Charger (Thales)	100723-513	TBD	23386

30. Surface friction (4.7.1.8 – New Paragraph)
The following test shall be performed. Measure and report the coefficient of static friction per ASTM G115-10, Standard Guide for Measuring and Reporting Friction Coefficients.

31. Capacity discharge (4.7.2.3).
Apply a pulse load of 150 Watts for 2 milliseconds followed by 17.6W for 98 milliseconds, cycled continuously to final voltage.

32. Cycle life test (4.7.2.4).
Delete 4.7.2.4 from MIL-PRF-32383 in its entirety and replace with the following:

Cycle life test. The following test shall be performed.

a. Cycles 1 through 27 - Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Allow batteries to rest for not greater than one hour. Discharge at 18W to cutoff voltage. Allow batteries to rest for not greater than one hour prior to the start of the next cycle. Every seventh cycle a rest of not greater than five hours is permited before and after charge.

c. Cycle 28 - Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Perform the electric leakage test of 4.7.4.7d. Allow batteries to rest for not less than 4 hours. Discharge at 2A to cutoff voltage. Record battery capacity.

d. Repeat steps a through c above until the 224 cycles are obtained.

e. After completion of the cycle life test, visually examine batteries for the criteria of TABLE VIII. Batteries shall meet the cycle life and visual mechanical requirements (see 3.1 and 3.5.3).

33. Pulse discharge (4.7.2.11).
Discharge with 5 amperes for 5 seconds, rest for 25 seconds. Continue discharge to final voltage to verify short circuit protection.

34. State of charge indicator (4.7.2.14).
Delete 4.7.2.14 from MIL-PRF-32383 in its entirety and replace with the following:

4.7.2.14 State of charge indicator. The following test shall be performed.

a. Charge each battery in accordance with 4.6; use of 4.6.3 is permitted.

b. Verify location and marking.

c. c. Press and hold the SOCI actuator. Measure and record the following to the nearest tenth of the second: (1) time from actuation to full intensity of the lights; (2) time from reaching full intensity until the lights extinguish; and (3) the total time from actuation until the lights extinguish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

d. Measure displacement and resistance values for the actuator and record the results.

e. Press each sample against a flat, transparent surface with not less than 5 lbs of force, with the face including the actuator against the surface.

f. Observe whether or not the LEDs illuminate; record the response.

g. Discharge batteries at the C/10 rate as follows: one battery at −10 ± 2ºC (+14ºF) and one at 55 ± 2ºC (130ºF).

h. Continuously discharge batteries to the following approximate states of charge, stopping the discharge at each state: 90%; 70%; 50%; 30%; 10%.

i. Obtain the SOCI readings while the test battery is not subject to discharge. Observe the indication of the state of charge indicator, read the SMBus data output, and record both indications after each partial discharge. Environmental chambers may be paused or shut off while accessing batteries to obtain the SOCI readings. For samples tested at −10°C, care is needed to limit the freezing of condensation at the battery to test circuit interface.

j. Once the SOCI readings have been obtained, restart the environmental chamber and restart the discharge in not less than 15 minutes after return to the required test temperature (for samples tested at −10°C, allow sufficient time to clear the frost if condensation has frozen on batteries).

k. Discharge the batteries to final voltage.

l. Observe the indication of the state of charge indicator, read the SMBus data output, and record the indications. Batteries shall meet the requirements of 3.5.8."

35. SMBus (4.7.2.15).
The cable used to verify SMBus shall not be less than 48 inches.

36. Vibration (4.7.3.7):
Add the following:
Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Batteries shall be subjected to loose cargo testing in accordance with MIL-STD-810 Method 514 Procedure II for not less than 20 minutes duration. Batteries shall be initially be placed stacked one on top of another in the package tester with no other equipment present. Batteries shall meet the requirements of 3.6, 3.6.5, and 3.4.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

37. Immersion, shallow (4.7.3.8):
Delete 4.7.3.8 from MIL-PRF-32383 in its entirety and replace with the following:

4.7.3.5 Immersion, shallow. Batteries shall be tested as follows. Weigh batteries and record their weight. The test samples shall be configured as follows: half in operating configuration (with sealed mating connector attached) and half with dust cap removed.

Immerse the batteries to a depth of not less than 1 meter in salt water substitute conforming to ASTM D-1141 for not less than 2 hours. Prior to the start of the test, the battery temperature shall be no less than 27°C (81°F) above the water temperature and the battery shall be at a full state of charge. Install a sealed mating connector to the battery connector. The battery shall be completely immersed, in an upright orientation. Once immersed, the batteries in operating configuration shall be loaded with a constant 18-watt load and shall remain loaded for the duration of the immersion. After immersion, remove the batteries and dry exterior surfaces. Re-weigh batteries and visually examine batteries for the criteria of TABLE VII. Batteries shall meet the requirements of 3.6, 3.65, and the applicable specification sheet (see 3.1). For conformance testing (see 4.5.2), deionized water may be substituted for salt water.

38. *Transit drop, severe (4.7.3.9.2)*
Delete 4.7.3.9.2 from MIL-PRF-32383 in its enterety and replace with the following:

4.7.3.9.2 Transit drop, severe. The following test shall be performed. Stabilize five of the batteries for not less than 4 hours at 55 ± 3ºC (130ºF) and the remaining five batteries for not less than 4 hours at -29 ± 3ºC (-20ºF). Perform the transit drop test per MIL-STD-810, Method 516, Procedure IV on each set of five batteries. Perform the first requisite drop per battery at ambient conditions in not greater than 10 minutes after removal from the preconditioning temperature, and the final requisite drop not greater than 30 minutes after removal from the preconditioning temperature. If greater than 30 minutes are required, stabilize the battery for an additional 4 hours before proceeding. Subject the batteries to the battery voltage tests of 4.7.2.2 and the electrical leakage test of 4.7.4.7d. After test, batteries shall meet the requirements of 3.6 and 3.6.7.

39. Impact (4.7.3.11 – New paragraph).
The following test shall be performed.
    a. Verify each battery meets the voltage requirements 3.5.2 and that the State of Charge Indicators are properly functioning.

    b. Place the battery on an ESAPI side plate backed by a sandbag.

    c. Drop a 16 lb Professional Bowlers Association (PBA) certified bowling ball or equivalent from not less than 36 inches onto the center-point of the battery +/- one inch.

    d. Repeat step 39 for the left and right sides of the horizontal centerlines of the battery not less than on half inch and not greater than two inches from the edge.

    e. Perform the capacity discharge of 4.3.7.2.

    Batteries shall meet the requirements of 3.5.3, 3.6, and 3.6.9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

40. *Conformability (4.7.3.12).*
The following test shall be performed.

a. *Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.*

b. *Soak the battery for not less than four hours at not greater than -20 ° C.*

c. *Place the battery between 2 curved fixtures, each having a radius of not greater than 15 inches oriented so that the vertical centerline of the battery makes contact with the fixtures.*

d. *Flex the edges of battery up and down until they make contact with the fixtures 50 times. (One flex consists of an upward and a downward motion)*

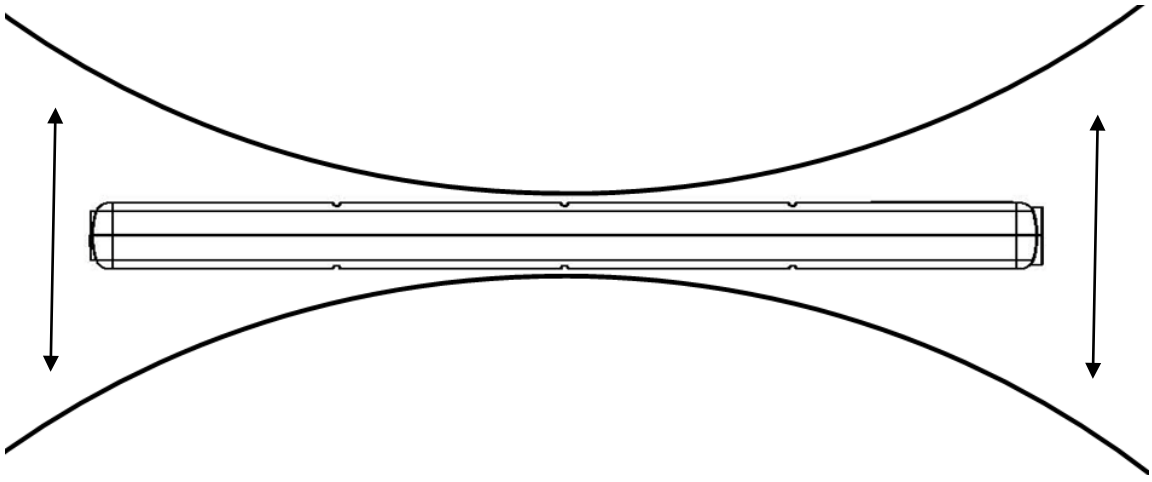
e. *Soak the battery for not less than four hours at not less than 60 ° C and repeat steps c and d*

f. *Allow the battery to stabilize at ambient temperature and flex the edges of battery up and down until they make contact with the fixtures 400 times. (One flex consists of an upward and a downward motion)*

g. *Repeat steps c and d using curved fixtures having a radius of not greater than 10 inches oriented so that the horizontal centerline of the battery makes contact with the fixtures.*

h. *Verify proper SOC functionality. Discharge batteries in accordance with 4.7.2.3c then perform the capacity discharge test of 4.7.2.3 using SMBus charging.*

Batteries shall meet the requirements of 3.5.3, 3.6, and 3.10



Note: Figure not to scale

FIGURE 5. Battery in conformability test fixture

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

41. *Explosive decompression (4.7.3.13 – New paragraph)*
The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Subject batteries to explosive decompression per Method 500, Procedure IV of MIL-STD-810. Half of the batteries shall be tested with a mating connector and half shall have an exposed connector (i.e. without dust cap). Perform the capacity discharge test of 4.7.2.3. Batteries shall meet the requirements of 3.5.3, 3.6, and 3.6.11.

42. *Chemical resistance (4.7.3.14 – New paragraph)*
The following test shall be performed. Batteries shall be sprayed with each of the following substances. The substance shall cover not less than 30% of the surface area of the battery and include portions of all seams, SOCI, vent, label, and charge contacts. After standing for not less than 10 minutes, the batteries shall be rinsed with soap and water. Batteries shall meet the requirements of 3.3.3, and TABLE VIII.
   a. *JP-8 per MIL-DTL-83133G*
   b. *Sodium Hypochlorite 5.25% (household bleach).*

43. *Solar radiation (4.7.3.15 – New paragraph)*
The following test shall be performed. Test batteries per MIL-STD-810, Method 505, Procedure II using the spectral power distribution below for seven days. Position one battery with the front face up and one battery with the front face down Wind speed shall be sufficient to maintain the battery surface temperature below 75 ° C. Below 65 ° C surface temperature wind speed shall be 0 mph. Batteries shall meet the requirements of 3.6.12 and 3.6.

Spectral Region	Bandwidth (nm)	Natural Radiation (% of total)	Tolerance (% of total)		Irradiance (W/m^2)	Spectral Region Irradiance (W/m^2)
			Min	Max		
Ultraviolet - B	280-320	0.5	0.3	0.7	5.6	5.6
Ultraviolet - A	320-360	2.4	1.8	3	26.9	62.7
	360-400	3.2	2.4	4.4	35.8	
Visible	400-520	17.9	16.1	19.7	200.5	580.2
	520-640	16.6	14.9	18.3	185.9	
	640-800	17.3	12.8	19	193.8	
Infrared	800-3000	42.1	33.7	50.5	471.5	471.5
				Totals	1120	1120

44. *Salt Fog(4.7.3.16 – New paragraph)*
The following test shall be performed. Test batteries per MIL-STD-810, Method 509 for four 24 hour cycles alternating 24-hour periods of salt fog exposure and drying conditions for a minimum of four 24-hour periods (two wet and two dry). The water used for the saltwater solution shall have a resistivity of not greater than 5 megohms per cm. The battery shall meet the requirements of 3.6, 3.6.15, and 3.5.3.

45. *Electrostatic Discharge (ESD) (4.7.3.17 New paragraph)*
The following test shall be performed. Subject the batteries to both the contact and air discharge test of IEC 61000-4-2, Level 4, to include each contact (both charge and main connector), connector shield, battery case, seams, and the SOCI. The battery

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

shall be tested with no load and under an 18W load. Verify that a load of 18W can be sustained for 1 minute following the test. Fully charge the battery via the SMBus and perform the discharge test of 4.7.2.3 once with the primary connector and once with the bottom charge contacts. The battery shall meet the requirements of 3.6, 3.6.13, and 3.5.3.

46. *Pulse magnetic field (ESD) (4.7.3.18 New paragraph)*
The following test shall be performed. Subject the batteries to the pulse magnetic field test of IEC 61000-4-9, Class 5, Severe Industrial Environment. The battery shall be tested with no load and under an 18W load. Verify that a load of 18W can be sustained for 1 minute following the test. Fully charge the battery via the SMBus and perform the discharge test of 4.7.2.3 once with the primary connector and once with the bottom charge contacts. The battery shall meet the requirements of 3.6, 3.6.14, and 3.5.3.

47. *Overcharge/electric leakage (4.7.4.7).*
Charge at 2.36 amperes with a charge voltage limit of 50 volts minimum during the 21-hour overcharge period of 4.7.4.7b. Discharge batteries at a constant 28W load to final voltage in step 4.7.4.7e. One sample shall be tested via the main connector and one sample shall be tested via the charge contacts.

48. *Bullet penetration(4.7.4.12)*
The following test shall be performed. Batteries shall be videoed at not less than 1080p 30 frames per second, and recorded by a thermal camera while subject to the following bullet penetration tests. Recordings shall not cease until the following conditions have been met: 1) 12 minutes following the impact; 2) violent activity (e.g. flames, rupturing) has ceased; 3) when the test sample is within 5 ° C of ambient.

a. *Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.*

b. *Mark a random location within each general area specified below to be shot.*

c. *Conduct each of the tests below targeting the spots marked in step b with not greater than 90 seconds between the 1st and 3rd rounds. Batteries shall be placed on a resistive load of 30 Ω +/- 10%. Muzzle velocity shall be 2390 FPS (+/- 800 FPS).*

d. *For each test record the date, time, place, ambient temperature, battery current (not less often than once per second), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity.*

Test A: Three (3) 7.62 mm API Tracer 45° Left Side Angle Impacts. The battery shall be shot three times at 45° angle from the front left face of the pack. Shoot the top left 1/3rd, followed by the central half, followed by the lower right section of the pack and observe and record test results.

Test B: Three (3) 7.62 mm API Tracer 90° Straight On Impacts. The battery shall be shot three times at 90° angle from the front face of the pack. Shoot the left mid 1/3rd, followed by the central half, followed by the lower right section of the pack and observe and record test results. An ESAPI or SAPI plate shall be present and the Conformal Battery shall be placed in an plate carrier pouch in front of the plate to emulate any

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

blowback from the plate behind the battery to ensure flame, fire, explosion does not occur.

49. Electromagnetic interference (4.7.7).
Delete 4.7.7 from MIL-PRF-32383 in its entirety and replace with the following:

4.7.7 Electromagnetic Interference (EMI) The following tests shall be performed. Batteries shall be tested while connected to the Nett Warrior and Air Soldier System or simulated systems. Range tests with radios and performance tests with navigation systems shall be evaluated and reported. Verify that the battery does not affect system functionality and test in accordance with the verification procedures of MIL-STD-461F in a lab environment while connected to the associated systems. As required, additional testing procedures are stated below.

RS103. Verify that the battery continues to power the associated system when the battery and associated cables are exposed to electric fields of 200 Volts per meter over the frequency range of 2 MHz to 40 GHz using the modulations shown Table 1 parts A and B of ADS-37A-PRF, in single channel receive mode. For the Nett Warrior application the on-channel input signal to the radio shall be -80 dBm or at a level of operation at just above the receiver sensitivity, whichever is more severe, for source signals (radiated energy) that are harmonics of the local oscillator plus or minus the IF (Intermediate Frequency). Compliance for this test shall be achieved with the radio and GPS antennas connected.

CE106. CE106 for the Nett Warrior configuration only shall be performed at the system level under the following configurations:
-NW System with Radio transmit & receive/GPS receive.
This test shall be performed for at least three fundamental frequencies for any radio used in a Nett Warrior configuration, and the two specified GPS frequencies.

Transmitter harmonics, except the second and third harmonic, and all other spurious emissions shall be at least 80 dB down from the level of the fundamental frequency. The second and third harmonics shall be suppressed 50 + log p dB (where p is the peak power output in watts at the fundamental) or 80 dB, whichever is less. For the Nett Warrior radio, three of the assigned frequencies shall be tested.

The CE106 data for both Nett Warrior and Air Soldier shall be recorded by a power spectral density plot for each radio equipment item.

RE102. For Air Soldier application and the 10 kHz to 18 GHz frequency range, use the limits of Figure RE102-3 of MIL-STD-461F (the curve for Fixed Wing External and Helicopters). Use the limits of Figure RE102-4 from 2 MHz to 18 GHz for the Nett Warrior application. The limits shall be adjusted for the receiver sensitivities over the frequency range of 420-450MHz. The requirement shall be met for both horizontally and vertically polarized fields above 30 MHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

System integration. In addition to the above testing, a system integration test shall be performed for Air Soldier equipment. The system shall be operated on a representative aircraft with a full suite of avionics systems. Avionics systems shall be tested in their worst (most likely to be affected by the battery) modes. Radios shall be tested at a minimum of 20 frequencies or 10 frequencies per operational band for any multi-band radio.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

50. TABLE IV

Delete TABLE IV from MIL-PRF-32383 in its entirety and replace with the following:

TABLE IV. First article tests, complete batteries

Inspection	Req't	Test Method	Samples or Group (GR)												
			GR1										GR2 (2 ea)	GR3 (8 ea)	GR4 (15 ea)
			1	2	3	4	5,6	7,8	9,10	11,12	13, 14	15, 16			
Visual and mechanical	3.4	4.7.1.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓	✓	✓							
Battery open circuit voltage	3.5.2	4.7.2.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Insulation resistance	3.7.2.1	4.7.4.6	✓	✓	✓	✓	✓	✓	✓	✓					
Altitude	3.6.2	4.7.3.4	✓	✓	✓	✓	✓	✓	✓	✓					
Explosive decompression	*3.6.2*	*4.7.3.4*	✓	✓	✓	✓	✓	✓	✓	✓					
Charge acceptance	3.5.10.1	4.7.2.9								✓					
Capacity discharge (initial)	3.5.3	4.7.2.3	✓	✓	✓	✓	✓	✓	✓	✓					
Cycle life	3.5.4	4.7.2.4					✓								
Battery storage life	3.6.8	4.7.3.10						✓							
Overcharge/electrical leakage	3.7.2.2	4.7.4.7							✓						
Low temperature discharge	3.5.3	4.7.2.7							✓						
High temperature discharge	3.5.3	4.7.2.8							✓						
Projectile	3.7.2.6	4.7.4.11							✓						
High rate discharge	3.5.3	4.7.2.5	✓		✓										
Retention of charge	3.5.10.2	4.7.2.10	✓	✓	✓	✓									
Pulse discharge	3.5.3	4.7.2.11	✓		✓										
Motor inrush current	3.5.5	4.7.2.12								✓					
Thermal shock	3.6.3	4.7.3.5	✓	✓	✓	✓									
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓									
Vibration (discharge)	3.6.5	4.7.3.7	✓	✓	✓	✓									
Conformability	3.6.10	4.7.3.13		✓	✓										
Connector insertion	3.4.4.1.1	4.7.1.3	✓			✓									
Flat terminal strength	3.4.4.1.2	4.7.1.4	✓			✓									
Immersion, shallow	3.6.6	4.7.3.8	✓	✓	✓	✓									
Transit drop, severe	3.6.7	4.7.3.9.2								✓				✓	
Surface friction	*3.4.5.6*	*4.7.1.8*									✓				
Salt Fog	*3.6.15*	*4.7.3.16*									✓				
Chemical resistance	*3.3.3*	*4.7.3.14*									✓	✓			
Electrostatic Discharge	*3.6.13*	*4.7.3.17*										✓			
Solar radiation	*3.6.12*	*4.7.3.15*									✓				
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1								✓	✓	✓		✓	
State of charge	3.5.8	4.7.2.14								✓					
SMBus	3.5.9	4.7.2.15	✓	✓	✓	✓									
Full capacity discharge	3.5.3	4.7.2.3	✓	✓	✓	✓									
Extreme low temp. discharge	3.5.3	4.7.3.2		✓		✓									
Extreme high temp. discharge	3.5.3	4.7.3.3	✓		✓										
Battery case vent	3.4.5.3	4.7.1.6	✓			✓									
Short circuit protection	3.7.2.3	4.7.4.8		✓	✓										
Impact	*3.6.9*	*4.7.3.11*		✓	✓										
High temperature temporary cut-off	3.7.2.4	4.7.4.9		✓	✓										
High temperature permanent cut-off	3.7.2.5	4.7.4.10		✓	✓										
Electromagnetic interference	3.8	4.7.7											✓		
Pulse magnetic field	3.6.14	4.7.3.18											✓		
Interchangeability	3.10	4.7.6												✓	
Lithium Battery Safety Program (US Navy) Tests	3.7.2.7	4.7.5													✓
Bullet Penetration	*3.7.2.8*	*4.7.4.12*								✓	✓	✓			

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

ADDITIONAL FIGURES

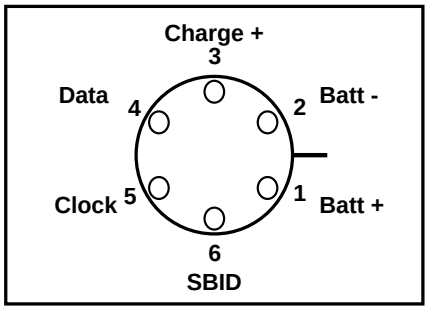
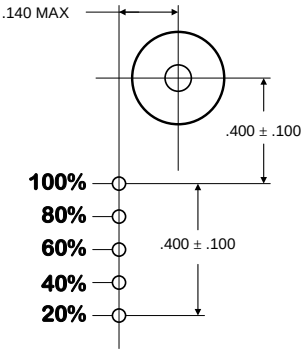
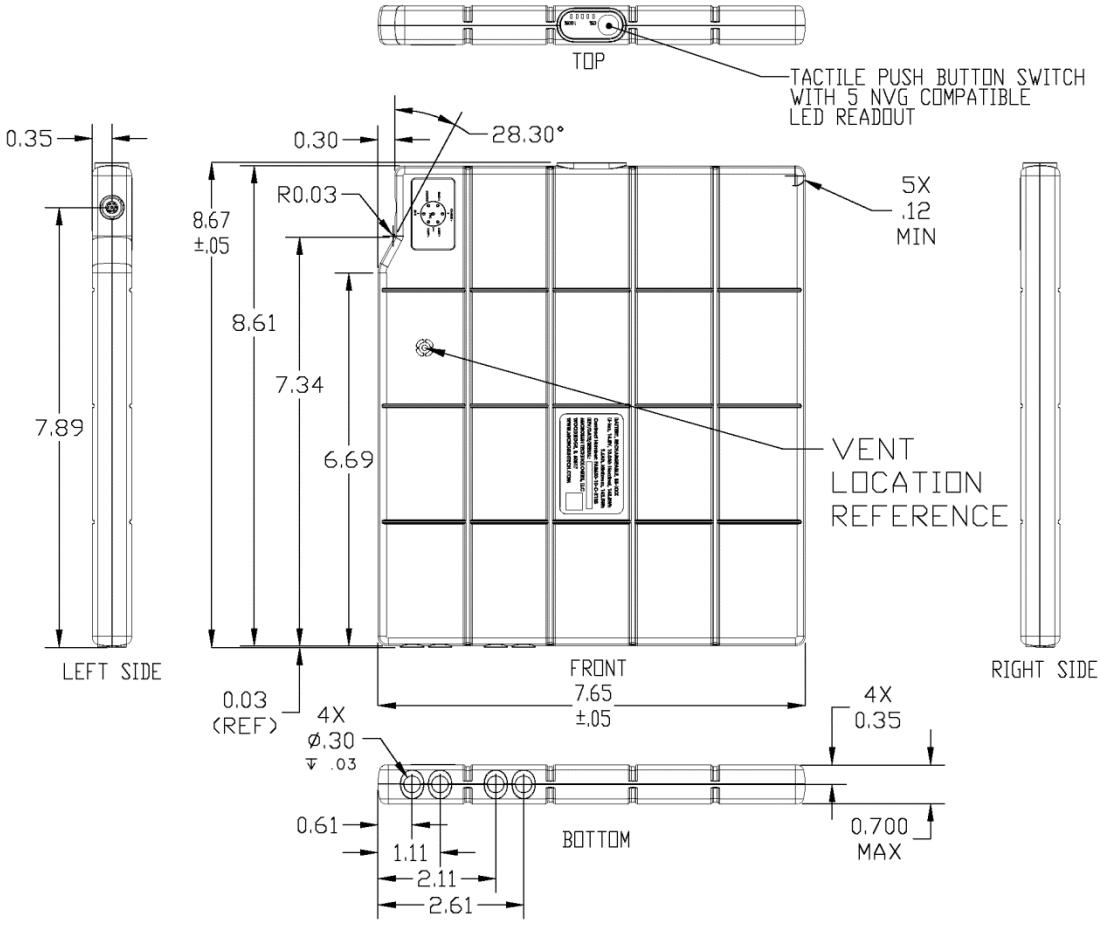


FIGURE 6. Terminal marking detail



NOTE: All dimensions are in inches.

FIGURE 7. State of charge indicator dimensions



NOTES: All dimensions are in inches. Tolerance for all is +/- 0.01 inches unless otherwise specified.

FIGURE 8. M32383/4-3 and M32383/4-3 Maximum Dimensions

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

NOTES: (This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This battery is intended for use in the Nett Warrior and Soldier Warrior systems. Please note that this battery must be electromagnetically compatible with these systems. Such compatibility is no guarantee of similar compatibility with any other system. Electromagnetic compatibility with any other system needs to be evaluated separately, and in the case of failure, either that system will need to adjust to this battery or another battery will need to be developed.

Electromagnetic interference testing. Verification requirements include electromagnetic interference tests of MIL-STD-461 with the battery connected to the system. Battery vendors will need to coordinate with the Project Manager, Soldier Warrior (SWAR), in order to arrange to have their products evaluated in the system. Their address is: Project Manager, Soldier Warrior, ATTN: SFAE-SDR-SWAR, 10125 Kingman Road, Building 317, Fort Belvoir, VA 22060-5820.

Bullet penetration. The Department of Defense has a preference for Bullet penetration testing to have Government safety personnel present to witness the testing. Potential offerors are cautioned to review solicitations for such requirements. Should this be required, battery vendors should contact Project Manager, Soldier Warrior (SWAR), in order to arrange to have their products tested. Their address is: Project Manager, Soldier Warrior, ATTN: SFAE-SDR-SWAR, 10125 Kingman Road, Building 317, Fort Belvoir, VA 22060-5820.

Below are some recommended safety precautions for bullet penetration testing:
Any conducted testing should have appropriate fire extinguisher, 5 gallon bucket of water or sand, safety glasses for personnel, oven type mitts when near test articles under test, appropriate ventilation or plan in case of thermal race conditions, and appropriate range safety personnel and procedures that are reviewed with all present in advance of the tests. Additionally fire log thongs or another device that is 32 inches or longer shall be present to move if needed an “active” system under test should it require it. Personnel performing weapons fire should be qualified on or in the direct presence of weapon qualified personnel. Range safety measures should be defined in advance and briefed to ALL personnel present during tests.

Other data:
Navy safety tests of NAVSEA S9310-AQ-SAF-010 and NAVSEA SG270-BV-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.

Nominal ratings: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes. The following is nominal data for the 150 Watt-Hour Variant:

Battery PIN:	**M32383/4-3 and M32383/4-4**
Type Classification:	BB-0150/U
Chemistry:	M32383/4-3 Li-Ion / M32383/4-4 Lithium Polymer
Color:	M32383/4-3 "Tan 686A" / M32383/4-4 "Earth Yellow"
Weight (max):	2.6 lbs (1180 grams)
Voltage Range:	10-16.8
Nominal Voltage:	14.8
Nominal Capacity:	10.0 Ah
Nominal Energy (new battery):	148 Wh at C/5
Battery Life:	≥ 224 cycles, ≥ 3 years
Rated power output:	≤ 148 W
Continuous load rating:	≤ 5.0 A
Pulse load rating:	35 A (915 μSEC) setting 30A ((≤25ms) Test
Charge temperature range:	-4 to 122ºF (-20 to 50ºC)
Operating temperature range:	-4 to 140ºF (-20 to 60ºC)
Storage temperature range:	-27 to 160ºF (-32 to 71ºC)
MAX abusive temperature (non-operating):	199ºF (93ºC)
Connector:	Glenair Part Number 8071-1472-12 / 8070-1299ZNU7-7DY - 807-663-12/807-348-01ZNU6-7SY equivalent
Overall Dimensions:	8.7 in. x 7.66 in. x 0.70 in.
Non-rechargeable equivalent:	N/A
Note 1: Maximum charge current for connector is 5A for connector and 5A bottom contacts	

Custodians:
Army – CR

Preparing activity:
Army – CR
(Project Number 6140-2011-010)

Review activities:
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T06:07Z
Check the source to verify that this is the current version before use.